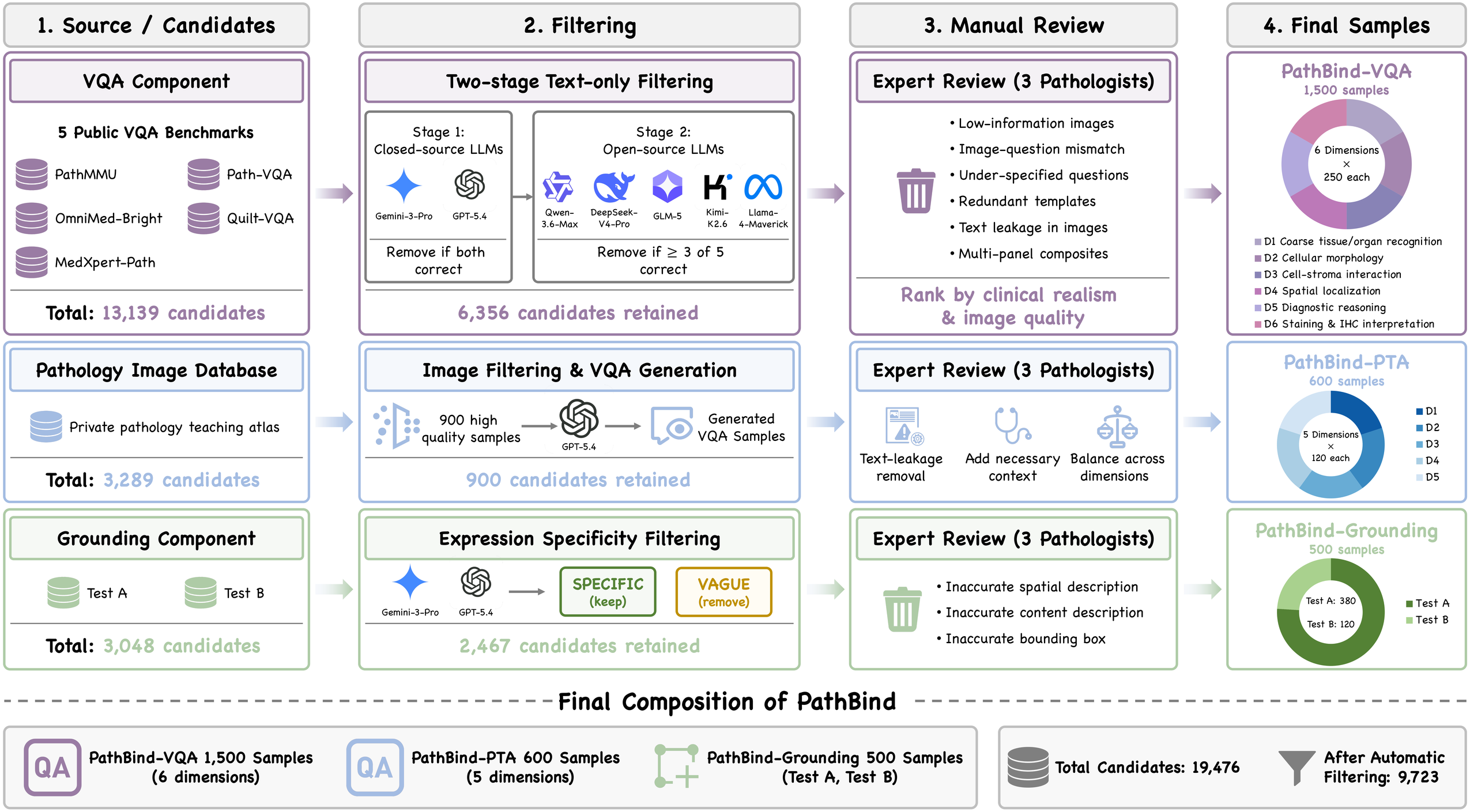

1. Source / Candidates
2. Filtering
3. Manual Review
4. Final Samples
VQA Component
5 Public VQA Benchmarks
PathMMU
Path-VQA
OmniMed-Bright
Quilt-VQA
MedXpert-Path
Total: 13,139 candidates
Two-stage Text-only Filtering
Stage 1: Closed-source LLMs
Gemini-3-Pro
GPT-5.4
Remove if both correct
Stage 2: Open-source LLMs
Qwen-3.6-Max
DeepSeek-V4-Pro
GLM-5
Kimi-K2.6
Llama-4-Maverick
Remove if ≥ 3 of 5 correct
6,356 candidates retained
Expert Review (3 Pathologists)
• Low-information images
• Image-question mismatch
• Under-specified questions
• Redundant templates
• Text leakage in images
• Multi-panel composites
Rank by clinical realism & image quality
PathBind-VQA
1,500 samples
6 Dimensions × 250 each
D1 Coarse tissue/organ recognition
D2 Cellular morphology
D3 Cell-stroma interaction
D4 Spatial localization
D5 Diagnostic reasoning
D6 Staining & IHC interpretation
Pathology Image Database
Private pathology teaching atlas
Total: 3,289 candidates
Image Filtering & VQA Generation
900 high quality samples
GPT-5.4
Generated VQA Samples
900 candidates retained
Expert Review (3 Pathologists)
Text-leakage removal
Add necessary context
Balance across dimensions
PathBind-PTA
600 samples
5 Dimensions × 120 each
D1
D2
D3
D4
D5
Grounding Component
Test A
Test B
Total: 3,048 candidates
Expression Specificity Filtering
Gemini-3-Pro
GPT-5.4
SPECIFIC (keep)
VAGUE (remove)
2,467 candidates retained
Expert Review (3 Pathologists)
• Inaccurate spatial description
• Inaccurate content description
• Inaccurate bounding box
PathBind-Grounding
500 samples
Test A: 380
Test B: 120
Test A
Test B
Final Composition of PathBind
QA
PathBind-VQA 1,500 Samples (6 dimensions)
QA
PathBind-PTA 600 Samples (5 dimensions)
PathBind-Grounding 500 Samples (Test A, Test B)
Total Candidates: 19,476
After Automatic Filtering: 9,723